



Mean

A typical value ^{or central value} is an amount that represents the values in a set of numbers. A typical value must fall within the series of numbers it represents. There are three common typical values in a set of numbers: mean, median, and mode.

When most people use the word average, they are talking about the mean. For a set of values that are similar to each other, the mean is what you want to know when you ask, “About how much...?”

To find the mean, add the values in a set of data, and then divide the sum by the number of values in the set.

Example 1

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Ella scored 14, 18, 11, and 16 points in her last 4 basketball games. What is her scoring average?

Step 1

Find the mean. Add the values in the set.

14 + 18 + 11 + 16 = 59 ^{aggregate score}

Step 2

Divide the sum by the number of values in the set.

$\frac{59}{4} = 14\frac{3}{4} = 14.75$

Ella’s scoring average is 14.75 points.

The mean is not necessarily equal to any number in the set. However, it must fall within the range of the numbers in the set.

If you know the mean, you can use algebra to find a missing value. You can also use the mean to draw conclusions and make predictions.

Example 2

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Sean budgets \$400 a month for food. In 5 months, he spent \$415, \$380, \$489, \$392, and \$404. In order to average \$400 per month, how much should he spend in the sixth month for food?

Step 1

Write an equation to represent the mean. Let x represent the amount spent in the sixth month.

$\frac{415+380+489+392+404+x}{6} = 400$

Step 2

Solve for x.

$\frac{415+380+489+392+404+x}{6} = 400$

$\frac{2,080+x}{6} = 400$

$2,080 + x = 2,400$
 $x = 320$

$6 \times 400 = 2,400$
 $x = 2,400 - 2,080$
 $x = 320$

Sean should spend \$320 in the sixth month to average \$400 per month.

Example 3

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Gina is paid weekly based on the amount of sales she makes. Her first paychecks are for the following amounts: \$346, \$358, \$320, and \$336. Based on the mean for the first 4 weeks, about how much will Gina make in a year?

Step 1

Find the mean of the first 4 weeks.

$\frac{346+358+320+336}{4} = 340$

Step 2

Gina will typically make \$340 a week. There are 52 weeks in a year. Multiply the mean by 52.

$\$340 \times 52 = \$17,680$

Gina will make about \$17,680 in a year.

Key Terms

- mean**
- median**
- mode**
- outlier**
- weighted average**

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Lesson 3: Averages

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